

Exploratory Data Analysis Report

E-Commerce Orders Dataset

Data Analytics Assignment II

Introduction

This report presents an exploratory data analysis (EDA) of a retail order dataset containing 1,200 transactions recorded between January 2023 and June 2025. The goal of this analysis is to understand the underlying patterns, trends, and distributions in the data, calculate key descriptive statistics, identify anomalies or outliers, and summarize observations that could inform business decisions.

The dataset captures 14 usable fields per order, including order identifiers, dates, customer IDs, product type, quantity, unit price, total price, shipping address, payment method, order status, tracking number, items in cart, coupon code, and referral source.

Dataset Overview

ATTRIBUTE	DETAIL
NUMBER OF RECORDS	1,200 orders
NUMBER OF FIELDS	14 (after removing empty helper columns)
DATE RANGE	1 Jan 2023 – 30 Jun 2025
MISSING VALUES	None detected across all fields
DUPLICATE ORDER IDS	None detected
PRODUCT CATEGORIES	7 (Chair, Desk, Laptop, Monitor, Phone, Printer, Tablet)

The dataset is clean and complete: there are no missing values or duplicate records, so no imputation or deduplication was required prior to analysis.

Descriptive Statistics

Basic descriptive statistics were calculated for the four numeric fields: order quantity, unit price, items in cart, and total order value.

Metric	Quantity	Unit Price (\$)	Items in Cart	Total Price (\$)
Count	1,200	1,200	1,200	1,200
Mean	2.95	356.41	5.49	1,053.97

Median	3.00	364.21	5.00	823.62
Std. Dev.	1.41	197.18	2.28	819.86
Minimum	1	11.39	1	11.39
Maximum	5	699.93	10	3,456.40

Order quantity is fairly uniform, averaging just under 3 units per order (range 1-5), suggesting quantity may have been generated within a fixed band rather than reflecting typical purchasing behavior.

Unit price ranges widely from \$11.39 to \$699.93, with a mean (\$356.41) close to the median (\$364.21), indicating a roughly symmetric price distribution across product categories.

Total order value shows a mean (\$1,053.97) noticeably higher than the median (\$823.62), a gap of about 22%, indicating the distribution is right-skewed a relatively small number of high-value orders pull the average upward. This is confirmed by a skewness coefficient of 0.89.

Items in cart average 5.49 (median 5), with a moderate spread (std. dev. 2.28) across a 1-10 range.

Trends

Order Volume and Revenue Over Time

Orders were placed steadily across the 30-month period, averaging roughly 40 orders per month with no single dominant seasonal peak. Monthly order counts ranged from a low of 27 (January 2025) to a high of 53 (June 2024). Monthly revenue tracked a similar pattern, fluctuating between roughly \$27,700 and \$68,100, largely in proportion to order volume rather than any sustained upward or downward trend revenue in mid-2025 sits close to revenue in early 2023, suggesting a mature, stable rather than growing or declining order stream over the period.

Product Category Performance

Product	Orders	Avg. Order Value (\$)	Total Revenue (\$)
Chair	178	1,098.99	195,620.11
Printer	181	1,080.73	195,612.61
Laptop	173	1,110.56	192,126.56
Tablet	179	1,042.28	186,568.95
Monitor	163	1,077.62	175,651.41
Desk	170	985.06	167,459.93
Phone	156	972.58	151,722.39

Revenue is spread fairly evenly across the seven product categories, with Chairs and Printers generating the highest total revenue (\$195,600 each) and Phones the lowest (\$151,700). No single category dominates, indicating a balanced product mix rather than reliance on one core product line.

4.3 Relationships Between Variables

A correlation analysis of the numeric fields shows that total price is most strongly associated with unit price ($r \approx 0.72$) and order quantity ($r \approx 0.62$), which is expected since total price is derived from these fields. Items in cart correlates moderately with quantity ($r \approx 0.65$) but only weakly with total price ($r \approx 0.39$), suggesting cart size reflects browsing behavior more than final purchase value. Unit price and quantity are essentially uncorrelated ($r \approx 0.01$), implying customers do not systematically buy more units of cheaper items.

Order Status and Payment Method

Order status is distributed almost evenly across five categories: Cancelled (250), Returned (247), Pending (237), Shipped (235), and Delivered (231) meaning only about 19% of orders in the dataset were successfully delivered. Payment method is likewise balanced across Online, Cash, Credit Card, Debit Card, and Gift Card (roughly 230-260 orders each), and no combination of payment method and order status stands out as disproportionately common, indicating no strong relationship between how customers pay and what happens to their order.

Outlier Analysis

Outliers were identified using the interquartile range (IQR) method (values beyond $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$). No outliers were found in unit price, quantity, or items-in-cart. Eight orders (0.7% of the dataset) were flagged as high-value outliers in total price, all exceeding \$3,330:

Order ID	Product	Qty	Unit Price (\$)	Total Price (\$)	Status
ORD200789	Tablet	5	691.28	3,456.40	Delivered
ORD201122	Monitor	5	678.19	3,390.95	Returned
ORD200632	Laptop	5	678.16	3,390.80	Delivered
ORD200469	Chair	5	676.98	3,384.90	Cancelled
ORD200328	Tablet	5	674.04	3,370.20	Cancelled
ORD200107	Printer	5	670.75	3,353.75	Shipped
ORD200326	Laptop	5	670.48	3,352.40	Returned
ORD201065	Printer	5	666.80	3,334.00	Delivered

All eight outliers share the same profile: maximum order quantity (5 units) combined with a near-maximum unit price. This is a coherent, explainable combination rather than a data-entry error, so these records were retained as valid extreme values rather than removed.

Key Findings

Total order value is right-skewed: the mean exceeds the median by about 22%, driven by a small cluster of high quantity × high unit-price orders rather than data quality issues.

Product category performance is well balanced, with total revenue per category ranging narrowly between \$151,700 and \$195,600 no single product dominates sales.

Order volume and revenue are stable across the 30-month period, with monthly fluctuations but no clear long-term growth or decline trend.

Only about 19% of orders reach 'Delivered' status, with Cancelled and Returned the two most common outcomes this is the most striking pattern in the data and would merit further investigation into fulfillment or data-generation processes.

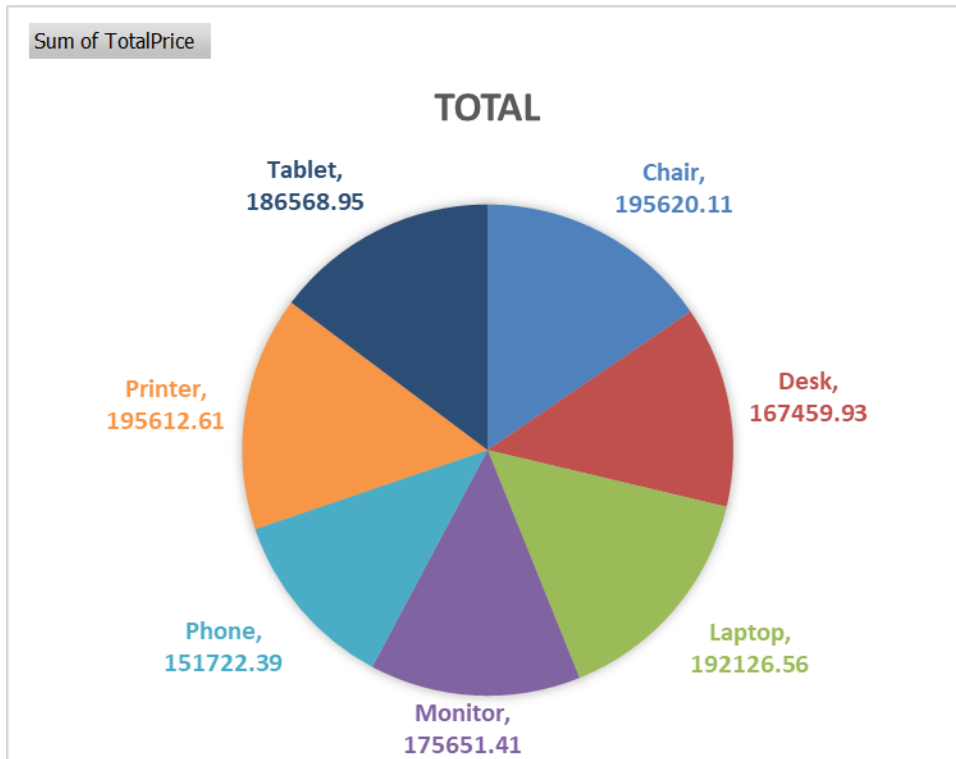
Payment method, referral source, and coupon usage are all evenly distributed, showing no strong bias toward any one channel or incentive.

Total price correlates most strongly with unit price and quantity (as expected structurally), while cart size behaves more independently, hinting it reflects browsing rather than purchase intent.

Conclusion

The exploratory analysis shows a clean, well-structured e-commerce dataset with balanced categorical distributions and a right-skewed total-price distribution driven by a small number of legitimately high-value orders. The most notable pattern is the low delivery-completion rate (19%), which stands out against otherwise evenly distributed order statuses and warrants further investigation in any follow-up analysis. Overall, no data quality issues were found that would require cleaning beyond what has already been applied in the dataset's 'Clean Dataset' sheet.

PivortChart Used are the following:



The pie chart displays the total sales (Sum of Total Price) for seven product categories. The sales are relatively evenly distributed, with Printer generating the highest total sales of 195,612.61, closely followed by Chair with 195,620.11 and Laptop with 192,126.56. These three categories contribute the largest share of total revenue.

Tablet recorded 186,568.95, while monitor generated 175,651.41. Desk contributed 167,459.93, and Phone had the lowest total sales at 151,722.39.

Key Insights

- Highest Sales: Chair – 195,620.11
- Second Highest: Printer – 195,612.61
- Third Highest: Laptop – 192,126.56
- Lowest Sales: Phone – 151,722.39
- The differences in sales among most product categories are relatively small, indicating a balanced sales distribution across the product range, with only the Phone category noticeably trailing behind the others.



Top 10 Orders by Total Sales Summary

The chart presents the **Top 10 orders** ranked by their sales performance based on **Total Price**. Overall, the selected orders generated **very similar sales values**, indicating a fairly balanced distribution of revenue across the highest-performing orders. None of the orders significantly outperformed the others, suggesting that sales were evenly spread among the top transactions.

Among the displayed orders, **ORD200789** recorded the highest total sales, while **ORD200463** and **ORD201031** were among the lowest within the top 10. Despite these slight differences, the variation in sales is minimal, reflecting consistent order values across the dataset.

This analysis indicates that the business does not rely heavily on a single high-value order for revenue generation. Instead, revenue is distributed across multiple orders, demonstrating stable sales performance. Such insights can help management monitor top-performing transactions, identify valuable customers, and maintain consistent sales strategies to sustain revenue growth.